

Mini Project Report - 11

Master of Computer Application – GENERATIVE AI
Semester – I

Sub: FRONT-END FRAMEWORKS AND TECHNOLOGIES

Topic: WEATHER INFORMATION APP

By
Name: R KUSHMA

Reg no.: PROV/ASAC/MCA/7/26/031

Faculty Name: VEERA RAGHAV K

Faculty Signature: _____

Department of Computer Application
Alliance University
Chandapura - Anekal Main Road, Anekal
Bengaluru - 562 106

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INTRODUCTION

The Weather Information App is a web-based mini project developed using HTML, CSS, and JavaScript. The main objective of this project is to create a simple, attractive, and user-friendly web application that allows users to enter a city name and view weather-related information. Weather information is useful in our daily lives because it helps people plan their activities, travel, and other outdoor tasks according to the weather conditions.

This project provides a basic understanding of how web technologies can be used to develop an interactive application.
In the present digital world, websites and web applications are widely used to provide information quickly and conveniently. A weather application is one such application that helps users understand different weather details in an organized manner.

The Weather Information App is designed with a simple interface where users can enter the name of a city in the input field and click the Search button to display the weather information for that city. The application displays details such as the city name, temperature, weather condition, humidity, and wind speed.

The project is developed using three main technologies: HTML, CSS, and JavaScript. HTML, which stands for HyperText Markup Language, is used to create the basic structure of the webpage. It defines the headings, input field, buttons, and sections where weather details are displayed.

CSS, which stands for Cascading Style Sheets, is used to improve the appearance of the webpage. It helps apply background colours, text styles, spacing, borders, rounded corners, and other design elements. In this project, a soft lavender background, a white content card, and purple buttons are used to create an attractive and comfortable user interface.

JavaScript is used to add functionality and interactivity to the application. It reads the city name entered by the user and checks whether the input field is empty. If the user clicks the Search button without entering a city name, an alert message is displayed asking the user to enter a city name. If a city name is entered, the application displays the city name along with sample weather details, including temperature, weather condition, humidity, and wind speed. This helps demonstrate the use of JavaScript functions, variables, conditional statements, and DOM manipulation.

Overall, the Weather Information App is a useful mini project for understanding the fundamentals of web development and creating a simple interactive application.

INPUT CODE

HTML CODE:

```
<html>
<head>
   <title>Weather Information App</title>
   <link rel="stylesheet" href="weather1.css">
</head>
<body>

   <div class="container">
      <h1>Weather Information App</h1>
      <p class="subtitle">Check your city's weather</p>

      <input type="text" id="city" placeholder="Enter city name">

      <button onclick="showWeather()">Search</button>

      <h2 id="cityName"></h2>
      <p id="temperature"></p>
      <p id="condition"></p>
      <p id="humidity"></p>
      <p id="wind"></p>
   </div>

   <script src="weather1.js"></script>
</body>
</html>
```

CSS CODE:

```
body {
   font-family: Arial, sans-serif;
   background: linear-gradient(to right, #d8c7ff, #f3d9e8);
   text-align: center;
   margin-top: 100px;
   min-height: 100vh;
}
```

```
.container {
    background-color: #ffffff;
    width: 350px;
    max-width: 85%;
    margin: auto;
    padding: 25px;
    border-radius: 20px;
    box-shadow: 0 8px 20px rgba(91, 57, 125, 0.2);
}

h1 {
    color: #6842a0;
    font-size: 26px;
}

.subtitle {
    color: #888099;
    font-size: 15px;
}

input {
    padding: 12px;
    width: 85%;
    box-sizing: border-box;
    border: 1px solid #cbb8e8;
    border-radius: 8px;
    margin-bottom: 12px;
}

button {
    padding: 12px 25px;
    background-color: #8055b8;
    color: white;
    border: none;
    border-radius: 8px;
    cursor: pointer;
}

button:hover {
    background-color: #603b91;
}
```

```
h2 {
   color: #6842a0;
   margin-top: 25px;
}

p {
   font-size: 17px;
   color: #514563;
}
```

JS CODE:

```
function showWeather() {
   var city = document.getElementById("city").value.trim();

   if (city == "") {
      alert("Please enter a city name");
   } else {
      document.getElementById("cityName").innerHTML =
         "City: " + city;

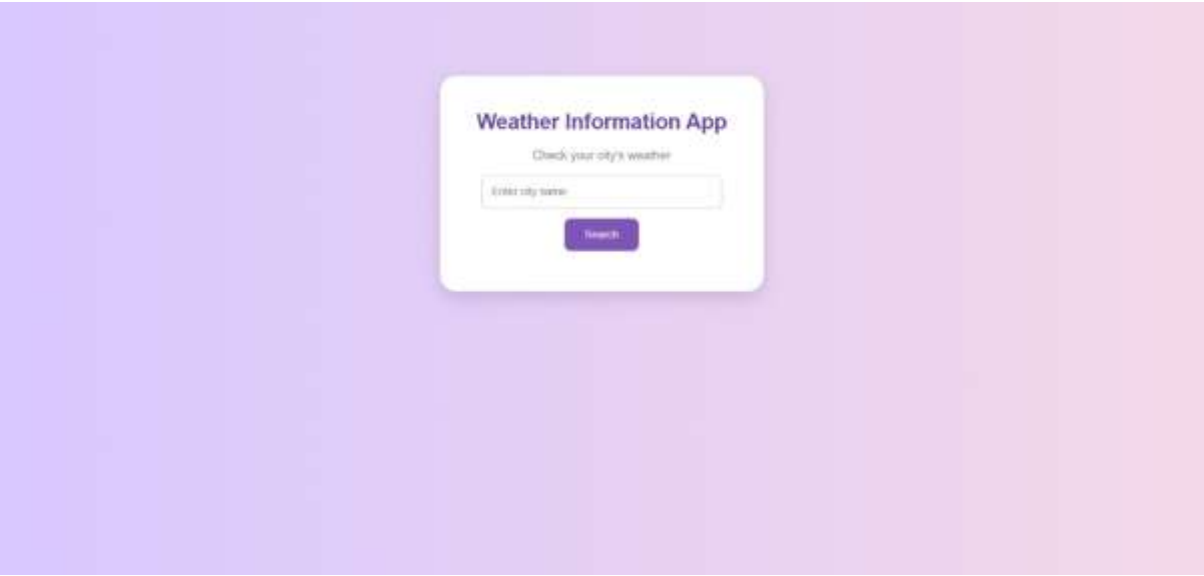
      document.getElementById("temperature").innerHTML =
         "Temperature: 26 °C";

      document.getElementById("condition").innerHTML =
         "Weather Condition: Cloudy";

      document.getElementById("humidity").innerHTML =
         "Humidity: 65%";

      document.getElementById("wind").innerHTML =
         "Wind Speed: 12 km/h";
   }
}
```

OUTPUT CODE


Weather Information App
Check your city's weather
Search


Weather Information App
Check your city's weather
bangalore
Search
City: bangalore
Temperature: 26 Â°C
Weather Condition: Cloudy
Humidity: 65%
Wind Speed: 12 km/h

CONCLUSION

The **Weather Information App** mini project was developed to understand the practical application of HTML, CSS, and JavaScript in creating a simple and interactive web application. The project demonstrates how a webpage can be designed to accept user input and display information in a clear and organized format. Through this project, the basic concepts of web development were applied to create an application with a user-friendly interface and simple functionality.

HTML was used to develop the structure of the application by creating the heading, input field, Search button, and sections for displaying weather information. CSS was used to improve the visual appearance of the webpage by applying suitable background colours, font styles, spacing, borders, and rounded corners. The lavender background, white content card, and purple buttons give the application a neat and attractive appearance. These design elements help make the webpage easy to understand and use.

JavaScript played an important role in making the application interactive. A JavaScript function was created to read the city name entered by the user and check whether the input field was empty. When the user clicks the Search button without entering a city name, an alert message is displayed. When a city name is provided, the application displays the city name and sample values for temperature, weather condition, humidity, and wind speed. This helped demonstrate how JavaScript can respond to user actions and update the content of an HTML webpage dynamically.

The project also has certain limitations. The current version displays predefined sample weather values instead of retrieving actual weather information from a weather service. Therefore, the temperature, humidity, wind speed, and weather condition shown by the application may not represent the real conditions of the selected city. However, the project successfully demonstrates the basic design and interaction required for a beginner-level weather application
.
In the future, the application can be improved by integrating a weather API to retrieve live weather information for different cities. Additional features such as weather icons, current location detection, a seven-day forecast, sunrise and sunset timings, and improved mobile responsiveness can also be included. Input validation and better error messages can further improve the user experience.